

ZTL — Zero-Trust Logic

V. Reznik. Preprint, v1.2 — 2026-07-13. Concept DOI: [10.5281/zenodo.21318981](https://doi.org/10.5281/zenodo.21318981) (v1.1: [10.5281/zenodo.21323552](https://doi.org/10.5281/zenodo.21323552); v1.0: [10.5281/zenodo.21318982](https://doi.org/10.5281/zenodo.21318982)). v1.2 adds: the central construction named — the zero-trust lift (§2); §3.8, an explicit Lean-verified census of the sixteen lifted binary connectives — re-deriving Finn’s external-Bochvar completeness landscape (Finn 1974): solo-completeness tracks non-commutative directionality (Sheffer’s stroke and Peirce’s arrow fall, both implications and both abjunctions survive), and the surviving basis $\Rightarrow$ reads as the credit detector; the fence-depth theorem in §19 (exactly $m-1$; no constant-depth fence); the at-scale stress-test of the warranty ladder; the recalculated Bochvar ledger (§4); and the three-laws-of-thought reading — a denial is free, an affirmation is on credit (§3.1); and the paradox engine (§11) — the expeditions unified as one construction $\text{paradox}(f)=\text{ground}(S=f(S))$ with three measured layers. The tag MEASURED means “verified by machine enumeration” (code in this repository), as opposed to “argued”; references to Lean mean proofs checked by the Lean 4 kernel with an **empty axiom list**.

Abstract

Six independent engineering traditions — IEEE 754 arithmetic (NaN), SQL’s three-valued logic (NULL), taint tracking in security, abstract interpretation in static analysis, imprecise probabilities in decision theory, and provenance semirings in database theory — have spent decades solving one and the same problem: how to compute honestly over unverified data. We show (MEASURED) that all six implement fragments of a single logic, and that this logic is generated in its entirety by one principle: **truth is never granted on credit** — a connective returns T only if T is forced under every classical reading of the unverified. There are exactly two truth values (verdicts are always classical); the third symbol Z is a mark on an unverified input, not a truth value.

For this logic (ZTL, Zero-Trust Logic) we build: a complete semantic account with a measured price list (12 surviving laws, including modus ponens, and 14 fallen ones — all the fallen laws are “truth from form”); the discovered split between *rules* and *laws*, with a one-directional deduction theorem for the primitive arrow; a signed tableau calculus with machine-proven soundness, completeness and cut admissibility; an algebraic passport (the fallen idempotence yields exact truth detectors, the external layer is expressively complete, a definable external implication restores the full deduction theorem, Craig interpolation holds, and the Blok–Pigozzi conditions are verified on the matrix — ZTL is algebraizable, yet not self-extensional); quantifiers; a modal identification (local modality over the S5 frame of completions — versus global supervaluation); a probabilistic identification (verdicts are the $\{0,1\}$ -threshold of Dempster–Shafer belief functions); a theory of verification (a verdict is a pair “value + warranty”, where the warranty is a two-grade ladder: sound — never lies; hereditary — never revoked) and of evidence combination (conflict is never renormalized — Zadeh’s paradox is resolved in Smets’ favor); and a quarantine passport that types every refusal by its genesis — paradox (permanent), intrinsic (the stipulation is forced), underdetermined (until a choice), unverified input (until verification), inherited — with a measured stipulation theorem separating the liftable from the permanent. The entire development — the core, both engine certificates with cut admissibility, the algebraic witnesses, the general fixed-point theorem and the expedition twins, twelve modules in all — is formalized in Lean 4 **with an empty axiom list, definitions included**. As a test bench the logic is run over the classical paradoxes — the liar, Jourdain’s carousel, Curry, Yablo, the crocodile, Russell — and in every case explosion is replaced by pointwise quarantine (for Russell, 8 of 9 membership facts stay grounded; the uncountability of the continuum splits into two independent failures). Functionally the $\{\neg, \wedge, \vee\}$ fragment coincides, cell by cell, with the external layer of Bochvar’s logic (1938) — a kinship found in the literature search after the tables had been generated, not a source; the contribution of this work is the generating principle, an implicational floor lying outside the Rosser–Turquette standardness

conditions, the calculus, the machine verification, and the bridges to the engineering traditions. An interactive studio ships with the repository: natural language is negotiated into ZFL (the Zero-trust Formal Language), a small language whose validity guarantees loadability, and judged by the measured core — the pipeline itself obeys the logic it serves (the LLM’s output is an unverified input; the deterministic core is the customs).

1. Motivation: one problem, six independent solutions

Most of the data over which modern software computes is unverified: sensor readings, user input, third-party databases, answers of network services. Engineering answered not with a single theory but with local inventions:

- IEEE 754 (1985): NaN — arithmetic is infected, comparisons refuse;
- SQL (1986): NULL — three-valued logic inside expressions, forced falsehood at the WHERE boundary;
- taint tracking (Denning, 1976; Perl taint mode, TaintDroid): a distrust mark flows through computations, and only an explicit check sanitizes;
- abstract interpretation (Cousot & Cousot, 1977): interval values flow, assertions are checked for being forced;
- imprecise probabilities (Walley, 1991) and Dempster–Shafer theory: ignorance is an interval [Bel, Pl], not a point probability;
- provenance semirings (Green–Karvounarakis–Tannen, 2007): trust in a fact is a polynomial over its sources.

Each of these inventions parried its own special case of one disease: naive treatment of the unverified manufactures confidence out of nothing. We show that the six practices share a common denominator — a two-valued logic over marked inputs with a single generating principle — and that this denominator survives a full logical development: a calculus, quantifiers, modal and probabilistic semantics, machine verification. Along the way the classical paradoxes of self-reference, from the liar to Russell, receive a uniform diagnosis (quarantine instead of explosion) — they serve as a test bench, not as the point of departure.

The principle from which everything is built borrows its name from security: default deny. A defective input may be granted neither classical value, but every compound assertion about it must receive a classical verdict — “true only if forced”.

2. Definitions

Truth values: T (earned truth), F (falsehood). **Input mark:** Z (zero-trust, “not earned”) — a property of an atomic datum, not a truth value; it participates in the calculating tables as a third symbol.

Definition (the zero-trust lift). For any classical connective f , its lift:

$f^*(x_1, \dots, x_n) = \bigwedge \{ f(v_1, \dots, v_n) : v_i \in \text{subs}(x_i) \},$
 where $\text{subs}(Z) = \{T, F\}$, $\text{subs}(v) = \{v\}$ otherwise;
 $\bigwedge$ is classical conjunction over all combinations.

Every occurrence of Z is substituted independently; the result is always classical. This one construction is the generating principle of the whole system: every connective of ZTL is a lifted classical one, and every result in this preprint is a property of the lift. It is NOT the strict (Kleene-style) lift of domain theory, which propagates the extra element ($f(\dots, \perp, \dots) = \perp$): the strict lift passes the mark on, the zero-trust lift interrogates it — every classical reading is taken and truth must be unanimous. Throughout the text and the companion repositories it is called simply *the lift*; its emblem is $\updownarrow$ — the lift raises the operation into the marked world, the greedy collapse returns the verdict to the classical floor: the mark never lives above the ground level. (In formulas the lift stays f^* ; $\updownarrow$ is heraldry, not notation.)

Corollary (greediness theorem, MEASURED): no compound formula ever takes the value Z; Z lives only on atoms.

Tables (generated by the principle; the anchor cells were postulated at design time and are reproduced by the principle — `ztl.py`):

x	$\neg x$	$\wedge$	T	F	Z	$\vee$	T	F	Z	$\rightarrow$	T	F	Z
T	F	T	T	F	F	T	T	T	T	T	T	F	F
F	T	F	F	F	F	F	T	F	F	F	T	T	T
Z	F	Z	F	F	F	Z	T	F	F	Z	T	F	F

$\oplus$ and $\leftrightarrow$: every cell involving Z equals F; on classical inputs — classical.

Generating basis (MEASURED): $\{\neg, \wedge, \vee\}$ generates everything: $p \rightarrow q = \neg p \vee q$, $p \oplus q = (p \wedge \neg q) \vee (\neg p \wedge q)$, $p \leftrightarrow q = (p \wedge q) \vee (\neg p \wedge \neg q)$ are surviving identities.

Entailment: $\Gamma \models \varphi$ iff every valuation making all premises T makes the conclusion T. Tarskian by construction.

3. Results (all MEASURED)

3.1 Laws: 12 alive, 14 fallen

Alive: MP, non-contradiction, transitivity of $\rightarrow$, commutativity and associativity of $\wedge/\vee$, both distributivities, and the three canonical definitions of the derived connectives. Fallen: $\neg\neg p = p$, both De Morgan laws, contraposition-as-identity, $\oplus = \neg(\leftrightarrow)$, idempotence of $\wedge/\vee$, absorption, the units ($p \wedge T = p$, $p \vee F = p$), excluded middle, $p \rightarrow p$, Peirce's law, $q \rightarrow (p \rightarrow q)$. The common trait of the fallen: each fails only on Z, and each is a law of “free truth” (truth from form or from polarity flip).

The three laws of thought (MEASURED). Of the classical triad, exactly one survives the lift. Non-contradiction $\neg(p \wedge \neg p)$ is T under every marking and hereditary (§19); identity $p \rightarrow p$ and excluded middle $p \vee \neg p$ both fall on Z. One reading covers the split: **a denial is free, an affirmation is on credit**. Non-contradiction denies — “not both” is forced ($p \wedge \neg p$ is F under every reading of a mark) without knowing p; identity and excluded middle affirm — “p implies p”, “p or $\neg p$ ” — and the lift will not affirm truth over an unverified p. The classical duality breaks along the very line the principle draws: non-contradiction and excluded middle are De Morgan twins, yet the lift keeps the one that withholds and drops the one that grants. The generating principle read on the oldest laws — truth is refused on credit, never granted on it.

3.2 The split between rules and laws

Of 14 classical inference rules taken as entailments, 12 survive — including contraposition-as-a-rule ($p \rightarrow q \models \neg q \rightarrow \neg p$) and the K-rule ($q \models p \rightarrow q$), whose law twins fell. Two fall: $\neg\neg$ -elimination ($\neg\neg p \models p$; Z leaks through double negation) and “tautology in the conclusion” ($p \models q \vee \neg q$; a fresh atom earns no truth). Classical logic cannot see the split because the deduction theorem glues it shut; in ZTL the deduction theorem works **left-to-right only**: $\models A \rightarrow B$ implies $A \models B$, but $p \models p$ while $\not\models p \rightarrow p$. The arrow is stricter than entailment: entailment transports earned truth, the arrow is a verdict that must earn its own. (The one-way-ness is a property of the primitive arrow, not of the language: a definable external implication satisfies the full deduction theorem — §3.6.)

3.3 Classification: paracompleteness

ZTL is paracomplete (LEM fell) but not paraconsistent in the $\models$ sense: $v(p)=v(\neg p)=T$ is unsatisfiable, there are no gluts, explosion is vacuously valid. Semantic MP is intact — ZTL is stronger in inference than Priest’s LP.

3.4 Paradoxes: quarantine, not tables

Negation has no fixed point ($\neg Z=F$) — the liar cannot be seated by the tables (MEASURED: enumeration of all values). Paradoxes are extinguished by the quarantine flag: Z-sentences are exempted from the Tarski schema. Quarantine is detectable from inside: $isZ(x) = \neg(x \leftrightarrow x)$. The same formula expresses the revenge liar; its content evaluates to T while the sentence is denied truth — a price paid deliberately (it is the standard price of all quarantine theories; here it is written out explicitly).

3.5 Incompatibilities (mini-theorems)

1. $\{\neg Z=F, T \rightarrow Z=F\} \Rightarrow$ contraposition-as-identity is impossible (verified by substitution).
2. A value housing the liar must be a fixed point of $\neg$; pessimism ($\neg Z=F$) excludes this $\Rightarrow$ the quarantine flag is irremovable.
3. Collapsing $Z \rightarrow F$ at the atom (instead of at the operator) restores all classical laws and turns the system verbatim into Bochvar’s isomorph B3 $\square$ — the novelty and the rules/laws split vanish.

3.6 The algebraic passport: completeness as a logic (MEASURED + Lean)

The fallen laws pay for the algebra. The chain, each link verified by total enumeration and kernel-checked in Lean (module `ZAlgebra`, empty axiom list):

1. **Fallen idempotence is a truth detector.** $J_T(p) = p \wedge p$ takes T exactly at $p=T$ (the decorrelated readings of Z kill the Z-diagonal); $J_F(p) = \neg p \wedge \neg p$ detects F; $J_Z = isZ$ detects Z. Three exact disjoint indicators — the Rosser–Turquette J-operators, grown from ZTL’s own connectives.
2. **Expressive completeness of the external layer.** Every external function $V^n \rightarrow \{T, F\}$ is a disjunction of indicator conjunctions over the cells of its table (MEASURED totally: 8 of 8 unary, 512 of 512 binary; the unary construction is a Lean theorem). The basis $\{\neg, \wedge, \vee\}$ generates the entire external clone.
3. **The full deduction theorem returns — one floor up.** The definable external implication $E(p, q) = \neg(p \wedge p) \vee (q \wedge q)$ satisfies $\Gamma, A \models B \Leftrightarrow \Gamma \models E(A, B)$ in both directions (MEASURED: 0 divergences on 324 triples where the primitive arrow diverges 20 times; Lean: `ddt_E` over the whole language, with premise lists). The logic does internalize its own consequence — with the meta-level “if A is true then B is true” made internal, not with the zero-trust arrow, whose one-way-ness (§3.2) stands.
4. **Algebraizability.** The same-value detector $\Delta(p, q) = (J_T p \wedge J_T q) \vee (J_F p \wedge J_F q) \vee (J_Z p \wedge J_Z q)$ and the truth equation $p \wedge p \approx \neg(p \wedge \neg p)$ (which holds exactly at $p=T$) witness the Blok–Pigozzi conditions: $\models \Delta(p, p)$; $p, \Delta(p, q) \models q$; congruence for all six connectives; $p \not\models \Delta(p \wedge p, \neg(p \wedge \neg p))$. All four verified on the matrix (MEASURED + Lean); by the Blok–Pigozzi characterization [31] **ZTL is algebraizable** — its equivalent algebraic semantics is the quasivariety generated by the three-element algebra. This is family kinship, not discovery: Bochvar’s external logic is algebraizable via the quasivariety of Bochvar algebras (Bonzio–Pra Baldi [32]); ZTL grows the witnesses from its own primitives, and we verify the conditions directly rather than inherit them.
5. **Yet not self-extensional.** $p \not\models p \wedge p$ (idempotence is dead as a law, alive as interderivability) while $\neg(p \wedge p) \not\models \neg p$ (at $p=Z$): interderivability is not a congruence. The same failure that breaks self-extensionality builds the detectors — one design decision seen from two sides.
6. **Structurality.** The substitution lemma (Lean: `evalF_subst`, by induction on the formula) gives: $\Gamma \models \varphi$ implies $\sigma\Gamma \models \sigma\varphi$ for every uniform substitution σ . Together with reflexivity, monotonicity and

cut (immediate from the definition), $\models$ is a **structural Tarskian consequence relation** — finitary and decidable, since the matrix is finite.

7. **Craig interpolation** holds, and its proof is one line on top of item 2: if $A \models B$, the projection of A onto the shared atoms (“some completion of A ’s private atoms makes A true”) is an external function of the shared atoms, hence a formula — and its J-DNF interpolates: $A \models I$ (A ’s own valuation is the witness), $I \models B$ (glue the A -witness to the current valuation; B does not read A ’s private atoms). MEASURED totally: 400 of 400 entailing pairs on the one-shared-atom pool, 32 of 32 on a two-atom cross-sample. The standard caveat: with an empty shared set the interpolant is a constant, and in the constant-free language $\mathcal{T} = \neg(x \wedge \neg x)$ spends a spare variable — exactly as classically.

3.7 The quasivariety, scouted (MEASURED)

A reconnaissance of the equivalent algebraic semantics — the quasivariety generated by $A = (\{T, F, Z\}; \text{the six connectives})$ — before committing to its structure theory:

- **Subalgebras:** exactly two — the Boolean core $\{T, F\}$ and A itself. C-extension, algebraically: the classical world is the unique proper subalgebra.
- **Congruences:** the measurement corrected our guess — A is NOT simple. The partition $\{T, F\} \parallel \{Z\}$ is a congruence, and it is exactly the *greediness kernel*: its compatibility IS the statement that no operation ever outputs Z . The congruence lattice is the three-chain $\Delta < \theta_Z < \nabla$, so **A is subdirectly irreducible with monolith θ_Z** — greediness is not merely a theorem about the algebra, it is its monolith. The quotient A/θ_Z is the two-element algebra in which every operation lands in the classical class: the mark evaporates, algebraically.
- **The clone theorem (exact):** the term operations of A are *precisely* the projections plus all external functions (unary $1 + 8$; binary $2 + 512$, by closure computation) — §3.6’s expressive completeness sharpened to a clone identity: nothing else sneaks in.
- **The Plonka probe (the structural discriminator):** Bochvar algebras [32] are built over Plonka sums, and Plonka sums of Boolean algebras satisfy every *regular* Boolean identity. A fails the regular identities $p \wedge p = p$, $p \vee p = p$, $\neg \neg p = p$ — so **A is not a Plonka sum of Boolean algebras**: the greedy quasivariety is structurally foreign to the weak-Kleene/BCA landscape. Kinship in expressive power, divorce in structure.

What a dedicated paper would need: a quasi-equational axiomatization of $Q(A)$, its subquasivariety lattice, and a representation theorem replacing Plonka sums. The reconnaissance says the ore is there; the mining is left as a separate work.

3.8 Single-operator completeness: the census of sixteen (MEASURED + Lean)

Zhegalkin’s classical theorem splits in two under the lift, and the halves part ways. The *basis* survives entirely: the clone generated by $\{\wedge, \oplus\}$ + constants equals the clone of the canonical basis $\{\neg, \wedge, \vee\}$ + constants — a kernel clone equality, with negation verbatim as $x \oplus T$ and the disjunction witness $J_T(x) \oplus y \oplus (x \wedge y)$, where the truth detector $J_T(x) = x \wedge x$ (the fallen idempotence law, §3.6) repairs the classical polynomial. The *ring* falls: the unit law $x \wedge T = x$, idempotence and the distributivity of $\wedge$ over $\oplus$ all die on Z (only the characteristic-2 law $x \oplus x = \perp$ survives — it is an anchor cell), so no multilinear canonical form exists; the normal-form role belongs to the J-DNF of the external layer (§3.6).

The census. Lifting each of the sixteen binary Boolean kernels and closing over projections and constants gives, totally:

fate	kernels	clone size
complete alone	$\rightarrow, \leftarrow, \leftrightarrow, \nleftrightarrow$	514
fallen	NAND, NOR	18 (one shared cage)
fallen	$\wedge, \vee$	7
fallen	$\oplus, \leftrightarrow$	258
degenerate	$p, q, \neg p, \neg q, \top, \perp$	4–8

The census law: a lifted binary connective is complete alone (with constants) iff its kernel is essentially binary and non-commutative. The four directional connectives survive — both implications (the classical arrow-and-falsum basis of Hilbert systems crosses the lift intact) and both abjunctions (nonimplication $p \wedge \neg q$ and its converse $\neg p \wedge q$, the *credit detector*: its single truth cell is “the consequent stands, the ground does not” — the very situation the logic is built to forbid). Every commutative kernel falls, including both classical one-operator champions: Sheffer’s stroke and Peirce’s arrow stall in the *same* eighteen-table cage, each still defining negation on its diagonal ($x \uparrow x = \neg x = x \downarrow x$) yet unable to rebuild its own De Morgan partner. Directionality, not any “mark-killing”, is the mechanism: the lifted implication mints forced truth over marked cells ($Z \rightarrow T = T$) and is complete regardless. Classical completeness theory is untouched — these are statements about the lifted operations relative to the lifted clone (for the classical landscape see Post; Martin [33]; Rosenberg [34]). The completeness landscape of the lifted clone itself, however, is *not* new: this clone is the external-Bochvar class B^3_{ex} , whose maximal (pre-complete) classes were determined by Finn [35, Remark 1] — the five lifted Post classes (preserving falsehood, preserving truth, self-dual, linear, monotone) together with **two additional maximal classes generated by the NOR- and NAND-type connectives**. A single binary kernel is complete alone in B^3_{ex} iff its lift lies in none of these seven; both classical champions fall precisely because each generates one of the two extra classes — the fact our census re-derives and machine-checks. Our contribution here is therefore not the completeness theory (Finn’s) but (i) the explicit kernel-verified census of all sixteen binary kernels with clone cardinalities below, (ii) the observation that solo-completeness tracks non-commutative directionality, and (iii) the reading of the surviving basis $\leftrightarrow$ as the credit detector. (Our census further finds the two champions sharing a *single* eighteen-table clone, which reconciles with Finn’s two extra classes exactly. There are precisely two external negations — the mark-intolerant J_0 (the mark reads as false) and the mark-tolerant one (the mark reads as true); Finn’s two extra maximal classes are their respective $\cap$ -clones (measured, `finn_reconcile.py`). Our greedy NOR is $J_0 x_1 \cap J_0 x_2$ cell-for-cell, and both zero-trust champions generate its class $B^3_{\text{ex}, \neg}$; the mark-tolerant negation generates the distinct $B^3_{\text{ex}, -}$. Zero-trust is therefore, at the clone level, exactly the choice of the mark-intolerant negation — the unverified reads as not-true, never as true — and $B^3_{\text{ex}, -}$ is the shadow it rejects. Matching Finn’s $\bar{x}$ to the mark-tolerant negation is forced by elimination; its verbatim definition is in [35].)

Machine verification (`lean/ZClone.lean`, the twelfth module, empty axiom list throughout): the negative half by the certificate method — the explicit 18-table cage, its closure under both champions a single `decide`, with Boolean membership by own recursion (the core decidability of list membership is propext-tainted); the positive half by witness terms spliced through table extensionality as data (nine cell equalities instead of `funext`), giving the kernel clone equalities $\text{clone}(\leftrightarrow) = \text{clone}(\{\neg, \wedge, \vee\}) = \text{clone}(\{\wedge, \oplus\})$, plus the seven-table cage banning lone $\wedge$ from negation. The only measured (not kernel) remainder is the cardinality of the common clone: $514 = 2$ projections + all 512 external binary tables, by exhaustive closure.

4. Place in the literature

The pedigree, with the ledger kept honestly. Nothing here was taken from Bochvar constructively: the tables were generated by the zero-trust principle (§2; the anchor cells were design axioms), and the kinship below

emerged only in the subsequent literature search — the repository’s commit history is the lab notebook of that order. What the search found: on $\{\neg, \wedge, \vee\}$ ZTL coincides verbatim with the external layer of Bochvar’s B3 (1938; built against the paradoxes) — $\neg = \top$ (the $\Diamond$ -negation), $\wedge = \cap$, $\vee = \cup$ — and the truth detector $J_T = p \wedge p$ of §3.6, derived here from the fallen idempotence, carries the table of Bochvar’s *primitive* assertion operator: what he postulated, the principle re-derives. There the coincidence ends. The delta is 7 cells in $\rightarrow, \leftrightarrow, \oplus$: a systematic closing of the places where Bochvar’s quarantine begets vacuous truth (his $\frac{1}{2} \supset 0 = 1$, our $Z \rightarrow F = F$). The internal dynamics are opposite — his meaninglessness is infectious (the internal layer), our mark evaporates at the first operator — and so is the ontology: his third symbol is a *value* sentences take; our verdicts are two-valued, and the non-classical letters are *marks* (Z on a datum, N in the solver, §10) — the alphabet is four letters, N, Z, F, T, in the genetic order — nothing (pending), the unsettled (a question with no answer yet), the free denial (default-deny), the earned affirmation (truth on a ground): nothing $\rightarrow$ doubt $\rightarrow$ default-no $\rightarrow$ grounded-yes. Z is born second, as raw doubt, and returns last, as the hardened liar (§10) — the pre-computer genesis, not the solver’s lifecycle. ZTL’s implication is his $\Diamond A \supset \Box B$ taken as a primitive; a polarity-adaptive translation ($\Box$ in positive positions, $\Diamond$ in negative ones) instead of a uniform one. ZTL falls outside Tomova’s classification of “natural implications” (it fails $p \rightarrow p$) and coincides with none of the literal paralogics of the Karpenko–Tomova lattice (2017). The *meaning* of the mark, as against its tables, has a separate and earlier ancestor: Łukasiewicz’s $\mathbb{L}_3$ [36] (1920), the first three-valued logic, whose middle value read *possible / not yet determined* — the nearest kin to Z’s *unverified until verification*. His motive was future contingents, not the paradoxes, and his implication keeps $p \rightarrow p$, so the kinship is of intent, not of tables. The two axes separate cleanly: the tables are Bochvar’s external B3, the mark’s meaning is Łukasiewicz’s, and no prior system occupied their conjunction. The passport’s two non-classical letters carry two distinct pedigrees accordingly — Z from Łukasiewicz’s indeterminate (a truth not yet settled), N from Kleene’s undefined (a computation not yet run, §10). Kindred in spirit: supervaluationism (rigid, non-tabular), subvaluationism/Jaskowski (its dual), SQL NULL (lazy), IEEE NaN (comparisons), exceptions with try/catch at every node (greediness). The motivation “greedy local supervaluation / default deny” was not found in the surveyed literature.

Devyatkin (2016, “Non-classical modifications of many-valued matrices”, read in full). The $\{\neg, \wedge, \vee\}$ core of ZTL fits the templates of his class 8Kb* — the paracomplete duals of the 8Kb family (Carnielli–Marcos, 8192 matrices): $\neg Z = F$, $Z \wedge x = F$, $Z \vee T = T$ land in the permitted “0 or 1” cells. So our triple is a member of a catalogued class — a stronger pedigree than previously known. However, the entire catalogue is built under the Rosser–Turquette standardness conditions for implication: middle $\rightarrow 0$ must be designated. ZTL with $Z \rightarrow F = F$ and $Z \rightarrow Z = F$ violates these conditions deliberately — the implication/equivalence floor of ZTL lies outside the catalogue, and it is precisely this floor that carries all the system’s signatures. The significance logics of Goddard–Routley (checked at survey level) use infectious nonsense plus classical external operators — Bochvar’s architecture, not per-operator collapse.

Database-theory neighbours. The closest kin is Libkin’s line: (a) certain answers (“true in every completion of an incomplete database”) = global supervaluation, the high-complexity theoretical gold standard (Libkin, ICDT’15/TODS’16); (b) Libkin–Peterfreund, “SQL Nulls and Two-Valued Logic” (PODS’23): SQL without the third value — atomic comparisons with NULL yield false (in both polarities, the same NaN asymmetry), while the Boolean layer is fully classical, restoring all laws (their goal is optimization). On the “where does indeterminacy collapse” scale, three of four positions are taken:

where the indeterminacy collapses	system
at the atom (predicate)	Libkin–Peterfreund 2VL $\approx$ Bochvar’s B3 $\Box$
at every operator	ZTL — position not found occupied
over the whole formula at once	certain answers / supervaluation
never (it flows)	Kleene / SQL 3VL inside expressions

The per-operator position is the only one that yields tabularity, two-valued verdicts and the internal signatures ($\neg\neg Z=T$, $Z\leftrightarrow Z=F$) at once; the price is the rewriting laws, which the atomic position preserves (which is why the optimizers chose it). A terminological caution: “local/global validity” in the philosophy of supervaluationism (McGee–McLaughlin, Varzi) is a distinction at the level of inferences, not of operators; our “locality” is a different notion.

5. The calculus: signed tableaux (MEASURED)

Hilbert style is closed off (axiom K fell, valid formulas are scarce), the deduction theorem is one-directional — the calculus is built as signed tableaux (the Rousseau–Hähnle architecture for finitely-valued logics).

Signs — four sets of values:

- strict: $\mathbf{T} = \{T\}$, $\mathbf{F} = \{F\}$;
- weak: $\mathbf{P} = \{T, Z\}$ (“possibly T”), $\mathbf{N} = \{F, Z\}$ (“did not earn T”).

Rules (branches separated by “|”, a comma puts both nodes on one branch):

$T : \neg\varphi \rightarrow F : \varphi$	$F : \neg\varphi \rightarrow P : \varphi$
$T : (\varphi \wedge \psi) \rightarrow T : \varphi, T : \psi$	$F : (\varphi \wedge \psi) \rightarrow N : \varphi \mid N : \psi$
$T : (\varphi \vee \psi) \rightarrow T : \varphi \mid T : \psi$	$F : (\varphi \vee \psi) \rightarrow N : \varphi, N : \psi$
$T : (\varphi \rightarrow \psi) \rightarrow F : \varphi \mid T : \psi$	$F : (\varphi \rightarrow \psi) \rightarrow P : \varphi, N : \psi$
$T : (\varphi \oplus \psi) \rightarrow T : \varphi, F : \psi \mid F : \varphi, T : \psi$	$F : (\varphi \oplus \psi) \rightarrow P : \varphi, P : \psi \mid N : \varphi, N : \psi$
$T : (\varphi \leftrightarrow \psi) \rightarrow T : \varphi, T : \psi \mid F : \varphi, F : \psi$	$F : (\varphi \leftrightarrow \psi) \rightarrow P : \varphi, N : \psi \mid N : \varphi, P : \psi$

On compound formulas $P \equiv T$ and $N \equiv F$ (the greediness theorem: a compound is never Z). **Branch closure:** the intersection of some formula’s signs is empty (T against N, F against P, T against F); a pair P and N on an atom does *not* close the branch — their intersection $\{Z\}$ yields a Z-countermodel. **Procedure:** $\Gamma \vdash \varphi \Leftrightarrow$ the tableau from $\{T:\gamma \mid \gamma \in \Gamma\} \cup \{N:\varphi\}$ closes entirely.

The zero-trust signature inside the calculus. T-polarity rules demand strict certificates (only T/F); weak signs appear exclusively in F-polarity. Classical tableaux are these same rules with $P \equiv T$, $N \equiv F$ glued — the whole contribution of Z is the unglueing of the negative signs. Proving truth in ZTL costs the same as classically; refuting is cheaper, because a countermodel may hide in Z.

Soundness and completeness (MEASURED): (a) each rule is machine-checked against the preimage of its table — the branches cover the preimage exactly; (b) the tableau decisions coincide with semantic $\models$ on 2462 entailments (the rule battery + all pairs of a generated formula pool + a sample of two-premise sequents). For finitely-valued tableaux with exact preimage coverage, soundness/completeness is the standard constructive result; the machine cross-check is an independent control of the implementation (and §8 upgrades it to a kernel-checked proof).

The sequent reading and semantic cut elimination. Read bottom-up, the tableau engine is a cut-free sequent calculus for the refutability judgment $\vdash S$ (“the signed set S is jointly unsatisfiable”): axioms are the contradictory atom constraints, the rules are the twelve signed expansions with premises above the line, derivability is the engine’s closure. The cut rule on a covering sign pair (every value lies in $\{T\} \cup \{F, Z\}$):

$$\frac{\vdash S, T:\varphi \quad \vdash S, N:\varphi}{\vdash S} \quad (\text{cut})$$

is sound, and the cut-free system is already complete — hence **cut is admissible**: the classic semantic cut elimination, here with a machine certificate. Kernel-checked with zero axioms (Lean: `cut_admissible`,

weakening_admissible, identity_refutable on top of closes_iff); MEASURED directly as well (identity 14/14; weakening 696 checks, 0 violations; cut 406 fired instances on each covering pair, 0 violations). What remains proof-theoretic future work is a *syntactic* cut-elimination procedure with complexity bounds — the admissibility itself is settled.

6. Quantifiers: finite domains and beyond (MEASURED)

By the generating principle: $\forall x\varphi = \mathbf{T}$ if every instance is strictly **T**, else **F** (one Z-witness poisons the universal); $\exists x\varphi = \mathbf{T}$ if some instance is strictly **T**, else **F** (a Z-candidate does not count as a witness). Greediness extends: quantified formulas never take Z.

Measured over all interpretations (unary P,Q — domains 1..3; binary R — domains 1..2):

- **Identities:** both distributions survive ($\forall$ over $\wedge$, $\exists$ over $\vee$); both quantifier De Morgan laws fell (counterexample: a one-element domain with $P=Z$ — Z hides under negation).
- **Instantiation asymmetry:** UI survives even as a **law** ($\models \forall yP(y) \rightarrow P(a)$ — the universal has earned its truth, spend it freely), EG as a law fell ($\not\models P(a) \rightarrow \exists yP(y)$ at $P(a)=Z$); only the EG rule survives.
- **The rules/laws split continues on the quantifier floor, and the first rule casualty appears:** $\neg\exists yP \not\models \forall y\neg P$ — from “no strict witness” it does not follow that “all are strictly false”: Z-elements remain. This is the quantifier twin of fallen $\neg\neg$ -elimination. The converse rule $\forall y\neg P \models \neg\exists yP$ survives, as does the quantifier swap $\exists x\forall yR \models \forall y\exists xR$.
- **Classical ornaments:** quantified LEM fell; the “drinker paradox” $\exists y(P(y) \rightarrow \forall zP(z))$ — a classical validity — fell in ZTL (in a bar with one Z-patron there is no drinker).

Quantifier tableaux (MEASURED): the finite-domain rules continue the sign signature — $T:\forall$ unfolds into strict $T:\varphi(a_i)$ on one branch, $F:\forall$ into weak $N:\varphi(a_i)$ across branches; $\exists$ mirrors. Tableau decisions coincide with semantic enumeration on 28 sequents (domains 1–2). A by-product: $\neg\forall yP \not\models \exists y\neg P$ even as a rule — the second fallen quantifier bridge, symmetric to $\neg\exists \not\models \forall\neg$ (negation hides Z in both directions). The finite-domain quantifier tableaux are now **kernel-checked** (Lean module ZQuant, zero axioms): over a finite domain the quantifiers are strict folds expressible in the certified language ($\forall$ as a conj-fold, $\exists$ as a disj-fold — on a singleton domain both collapse to the J_T guard $\varphi \wedge \varphi$ of §3.6), the n-ary signed rules are theorems about the folds (cover_allF_T/F, cover_exF_T/F), UI/EG hold in membership form over the whole language (ui_mem, eg_mem), and eight battery verdicts — including the failing drinker and quantified LEM — are kernel evaluations of the certified engine itself.

Arbitrary domains: parameter tableaux (MEASURED). Over arbitrary domains the finite unfolding is unavailable; the standard cure — parameter (free-variable) tableaux with γ/δ rules — carries the zero-trust sign discipline over exactly:

γ (reusable, every parameter):	$T:\forall x\varphi \rightarrow T:\varphi(c)$	$F:\exists x\varphi \rightarrow N:\varphi(c)$
δ (fresh parameter, once):	$F:\forall x\varphi \rightarrow N:\varphi(c^*)$	$T:\exists x\varphi \rightarrow T:\varphi(c^*)$

Fresh witnesses appear exactly where the propositional calculus allows weak signs (F-polarity) or demands a strict witness ($T:\exists$). Status, honestly split: **soundness is measured** — every sequent the engine proves is re-checked by total enumeration over finite domains, every saturated open branch yields a countermodel that is verified by evaluation (battery of 13: UI/EG, distribution, quantifier bridges, swap and its failing converse, the failing unguarded drinker and quantified LEM — all 13 verdicts confirmed). **Completeness is the standard Hintikka-saturation argument** for finitely-valued signed tableaux [27] — argued, not measured. Two honest FO phenomena appear on cue: on invalid sequents whose branches spawn witnesses forever (the unguarded drinker; the converse quantifier swap) the tableau does not terminate and invalidity is certified by a finite countermodel instead; and FO-ZTL is **undecidable** — the J-guard translation $P \mapsto P \wedge P$ (§3.6) makes every

atom classical, embedding classical first-order validity (the guarded drinker is ZTL-valid and needs the classic two γ -rounds). The tableaux give semi-decidability, as classically.

7. Limitations and honest caveats

- No new three-valued functions exist or are claimed (functional completeness — Finn); the contribution is the choice of primitives and the principle. Positively: the chosen primitives generate the whole external clone (§3.6), so ZTL is the external layer presented by different primitives — its abstract metatheory (algebraizability included) is a matter of verification, not invention.
- “Deviation from classical logic” is merchandise here, not defect: on classical inputs the deviation is zero (C-extension); all exotica is the price of the policy toward Z.
- The price list is not negotiable item by item: the fallen laws are consequences of three fixed design forks ($\neg Z=F$; greedy collapse; $Z \leftrightarrow Z=F$); regaining any law requires flipping a fork (see §3.5).

8. Machine verification in Lean 4

The core is ported to Lean 4 (v4.29.1, no mathlib): the connectives are generated by the lift (tables are not postulated — computed), the anchor cells are turned from postulates into theorems, the 12 alive and 14 fallen laws, semantic MP, the greediness theorem, the homelessness of the liar ($\forall v, \neg v \neq v$), the isZ detector and the quantifier UI/EG asymmetry — all proven.

Axiom status: the empty list. `#print axioms` over the whole corpus returns “does not depend on any axioms”: no Classical.choice, no Quot.sound, not even propext; pure computation. This is the strictest possible tier — rare for a substantial logical system.

The temporal modules (v1.2): `ZTime.lean` — the verification tree, with absorption, arrow and ladder-inclusion proven structurally for every formula and marking; `EpochBoundary.lean` — the epoch boundary theorem and the separation witness (§§21–22). Both self-contained, both on the empty axiom list, both verifiable from zero by a bare `lean` call.

Part II (same file): the entailment rule battery (11 alive + 2 fallen, including the split contraposition-rule vs contraposition-law), no-gluts, the lazy Kleene register with proven monotonicity of all connectives in the information order, non-monotonicity of the greedy register, the liar’s home in the lazy register (knot $Z = Z$ — rfl), the absence of a greedy carousel model (all 9 pairs), lazy grounding, and the computed revenge bullet.

Part III — the tableau pillars over the whole language. An inductive formula type `Fm` with evaluation; **pillar 1**: greediness is proven for the entire language (every compound formula is classical under every valuation — by constructor analysis, not by battery); **pillar 2**: the preimage coverage of each of the 12 tableau rules — as $\Leftrightarrow$ -theorems for arbitrary subformula values (`cover_*`).

The trade certificate (lean/TableauCert.lean). The tableau engine is formalized in full: working rules over the generating basis $\{\neg, \wedge, \vee\}$ (weak signs only in F-polarity), atoms handled by constraint intersection, the heavy connectives $\rightarrow, \oplus, \leftrightarrow$ reduced to the basis by the surviving identities (`imp_def/xor_def/xnor_def` — theorems of the core). Proven:

- `closes_iff` — **soundness and completeness**: the tableau closes $\Leftrightarrow$ the signed nodes are unsatisfiable (induction on weighted size, for all formulas);
- `tproves_iff` — **the entailment certificate**: $\Gamma \vdash \varphi$ by the engine $\Leftrightarrow$ every valuation making the premises T makes the conclusion T.

Six smoke runs of the certified engine coincide with the measured results ($\not\vdash p \rightarrow p$, $MP \vdash, \neg(p \wedge \neg p) \vdash, \not\vdash$ LEM, contraposition-rule $\vdash, \neg\neg$ -elimination $\not\vdash$). **The certificate’s axiom status: the empty list** — same

as the core. This was achieved by disinfecting every known source: structural recursion on fuel instead of well-founded recursion (the WF machinery pulls in `propext/Quot.sound`), signs as functions $V \rightarrow \text{Bool}$ instead of lists (Lean core’s list-membership decision procedure carries `propext`), a recursive satisfaction predicate instead of $\forall$ -membership, combinator chains of `Iff` instead of rewriting by equivalences (`rw` with an `Iff` applies `propext`), and hand-rolled `Nat` arithmetic instead of `omega` (`omega` carries `propext` and `Quot.sound`). **The native engine (TableauCertN.lean):** the engine with the native signed rules for $\rightarrow, \oplus, \leftrightarrow$ is certified by the same induction (`closesN_iff`), and the theorem `engines_agree` shows both engines return identical verdicts; the former footnote about their equivalence is closed by a theorem. Additionally (all with zero axioms): the Lean port of marked sets (§12: a mark earns membership nowhere; a marked set is not provably a subset of itself; $\{Z, Z\} \neq \{Z\}$; $|\{Z\}| = [1, 1]$) and a corpus of facts — domain-2 quantifiers ($\forall = \text{zand}$, $\exists = \text{zor}$: the UI law alive; EG, $\neg \exists \neq \forall \neg$, $\neg \forall \neq \exists \neg$, quantified LEM, the drinker — fallen) and dynamics (liar period 2, carousel period 4 with no fixed points, Curry homeless greedily and grounded lazily, cycle parity for lengths 2/3, the Yablo-3 truncation with a unique grounded model, the nullity of the crocodile’s deal). **The algebraic passport (ZAlgebra, zero axioms):** the J-indicators, unary expressive completeness as a single theorem over all 8 target tables, the full deduction theorem for E over the whole language with premise lists (`ddt_E`), all Blok–Pigozzi witnesses (Δ -spec, reflexivity, detachment, congruence for the six connectives, the truth equation, condition (iv)), the failure of self-extensionality, the substitution lemma and structurality of $\models$ (`entails_structural`). **The sequent reading (ZSequent, zero axioms):** cut admissibility on top of the engine certificate (`cut_admissible`), admissible weakening, derivable identity — the semantic cut elimination of §5, kernel-checked. **Quantifier tableaux (ZQuant, zero axioms):** finite-domain quantifiers as strict folds, the n-ary signed rules as preimage-coverage theorems, UI/EG in membership form, and the battery of eight tableau verdicts as kernel evaluations of the certified engine (§6). **General Knaster–Tarski (ZGround, zero axioms):** the lazy register over the whole language, monotonicity, the least fixed point by bounded iteration, and the absoluteness of the grounded part (§9). **Expedition twins (ZExped, zero axioms):** streams — the equality atom never earns T, one finite witness earns apartness and it persists, Cantor’s diagonal earns strict non-membership against every registry entry (§13); one marked pair collapses the injectivity certificate for every function including the identity (§14); interval decorrelation and unearned self-identity of a nondegenerate mark (§15); Dempster–Shafer thresholds (§16); atom verdicts as $\square/\diamond$ thresholds with the $\neg\neg$ -cell separating the local ladder from global supervaluation (§17). The certified language now carries the constants $T/\perp$ (the engines and both certificates extended, the sequent/quantifier/algebra modules unaffected), which closes the last expedition remainder: **Russell’s grounding half is kernel-computed** — `lfp RUSSELL = [F, F, T, F, T, F, F, F, Z]` by the certified iteration of §9: eight membership facts ground, exactly $R \in R$ stays quarantined (`russell_grounded`, `russell_verdicts`); $\vdash T, \perp \vdash \varphi$ and $\not\vdash \perp$ run through the certified engine. **The stitch (bridge.py):** one questionnaire, two engines — 141 kernel-computed answers (both registers cell by cell, the J-operators, E and Δ , certified-engine verdicts on a shared propositional and quantified battery with constants, lazy `lfp` of the zoo up to the nine-fact Russell system) compared mechanically against the Python stands on every regression run: zero divergences. Two further disinfection pitfalls surfaced here: an overlapping wildcard row in a match taints the DEFINITION itself with `propext` through the compiled matcher (invisible to theorem-level axiom prints — `kand/kor` were rewritten with explicit cells, and the corpus now prints definition-level axiom checks too), and core’s `List.length_map/length_replicate` are simp-proved and carry `propext` — replaced by hand-rolled inductions (likewise the core `Int` order lemmas, `omega`-proved: general interval statements live over `Nat`, `Int` stays for computation).

9. Quarantine as a fixed point: the two-register architecture

(MEASURED)

The quarantine flag of fork 2 in §3.5 is formalized à la Kripke. A system of sentences with a truth predicate

(λ : $\neg\text{Tr}(\lambda)$ etc.); the “jump” J re-evaluates the sentences under the current valuation; fixed points of J are self-consistent valuations. Two registers with DIFFERENT negations are put on trial: the greedy one (ZTL tables, verdicts: $\neg Z = F$ — “not earned”) and the lazy one (strong Kleene, the solver: $\neg Z = Z$ — “do not judge the uncomputed”; Z flows through connectives). What follows shows these are not two candidates for one role but two mandatory different roles.

Enumeration results (the zoo: liar, truth-teller, Jourdain’s carousel, the even cycle, a grounded chain, the avenger):

1. **The greedy jump is non-monotone** in the information order (witnesses found on every system) — the Knaster–Tarski argument does not apply to it.
2. **On odd cycles the greedy jump has no fixed points at all**; the iteration oscillates: liar — period 2 ($F \rightarrow T \rightarrow F \dots$), carousel — period 4 ($FF \rightarrow FT \rightarrow TT \rightarrow TF$), avenger — period 2. The oscillations of revision theory (Gupta–Belnap) arise here not as a postulate but as the behavior of the greedy iteration.
3. **The lazy jump is monotone** and has a least fixed point everywhere: grounded sentences receive classical values, paradoxical ones — Z . The even cycle: three fixed points ($\{T, F\}$, $\{F, T\}$, $\{Z, Z\}$), the least being mutual quarantine (underdetermination, not paradox).
4. **Verdicts are greedy, read over the finished point**: the content of the avenger μ at grounded $\mu = Z$ evaluates greedily to T , while μ is denied truth — the “bullet” is paid inside the formal construction, not by a disclaimer.

Conclusion (the architectural theorem of this stage): the two-register design is a necessity, not a convenience. The greedy register cannot ground itself (no fixed points on the liar); the lazy one cannot pass verdicts (“exactly falsehood”). Quarantine := the Z -set of the lazy jump’s least fixed point; ZTL := the greedy reading on top. The engineering precedent — SQL (Kleene inside expressions, forced falsehood at the WHERE boundary) — turns out to be not an analogy but the same theorem, found by practice.

The quarantine passport (MEASURED). Z alone is blind: the liar and the truth-teller land in quarantine with the same mark. The passport cures the blindness without touching the logic — verdicts, tables and greediness are intact; it is solver-side metadata computed per strongly connected component of the dependency graph. Kinds: **PARADOX** — the component has no classical model consistent with its grounded environment (odd cycles; the greedy oscillation period is recorded: liar 2, carousel 4) — the refusal is *permanent*; **INTRINSIC** — exactly one classical model exists (Kripke’s intrinsic value): ungrounded, yet uniquely consistent — the stipulation is forced, not chosen; **UNDERDETERMINED** — classical models exist (≥ 2 : even cycles, the truth-teller) — the refusal stands *until stipulation*; **INPUT** — a plain unverified datum — the refusal stands *until verification* (§19); **DOWNSTREAM** — inherited quarantine with the culprits listed (the provenance of refusal, §14 again). The operational content is the **stipulation theorem** (MEASURED totally on a mixed zoo carrying every kind at once, with the grounded part untouched): a component carries classical models (INTRINSIC or UNDERDETERMINED) iff stipulating any of them grounds it cleanly — the forced choice and the free one obey the same mechanics — and it is PARADOX iff every decree contradicts the component’s own definitions: the liftable and the permanent, mechanically separated. The passport is thereby a *biography* of the mark — there are exactly three ways to acquire it, and liftability follows genesis: **born** with the datum (INPUT — lifted by verification), **hardened** out of a solver phase that completed without resolving (INTRINSIC — lifted by the forced stipulation; UNDERDETERMINED — lifted by a chosen one; PARADOX — lifted by nothing), or **inherited** (DOWNSTREAM — lifted with the culprits). “Completed” is a theorem, not a hope: the lazy iteration provably terminates within $n+1$ steps (§8, Z_{Ground}), so the phase N always dies — the liar is never “still computing”; the solver’s verdict on it is final, and what is eternal is not the process but the refusal. (Contrast revision theory [7], where the process itself never settles.) The parity theorem of §11 re-derives through the passport (62 of 62 cycles), and Russell (§18) reads: $R \in R$ — PARADOX, permanent; the twin $S \in S$ — UNDERDETERMINED, awaiting an external decision; eight facts grounded. This is Kripke’s taxonomy

plus revision-theoretic signatures [5, 7], packaged as a computable instrument; the refusal classes now mirror §19, and quarantine = (Z, passport). Honest caveat: Yablo stays invisible — every finite truncation is grounded (§11), so the passport of infinite regress needs an infinite instrument.

The architecture, kernel-checked in general form. The two-register theorem no longer rests on per-instance measurements: the Lean module ZGround (zero axioms) proves, for EVERY finite system of definitions, that lazy evaluation is monotone over the whole language (evalK_mono), hence the lazy jump is monotone (jumpL_mono); the iteration from $\perp$ ascends and stabilizes within $n+1$ steps at the least fixed point (kt_fixed, via an information-measure pigeonhole — no classical choice, no well-founded machinery); the least point lies below every fixed point (kt_least); and a coordinate grounded in the least point carries the same classical value in every fixed point (grounded_absolute) — quarantine is well-defined, machine-checked. Together with the greedy register’s kernel-checked non-monotonicity (§8, Part II), the necessity of two registers is now a theorem end to end.

10. The ontological status of Z: the system’s passport

The final and most precise formulation of what has been built:

Truth values:	T, F	(verdicts are always two-valued)
Input mark:	Z "unverified"	(a property of data, not truth)
Solver state:	N "not yet computed"	(a computation phase, present only under self-reference;
		provably finite — §9 — and never escapes outward)
Reading policy:	local, default deny	(the three-symbol tables are the policy's calculator)

ZTL is a two-valued logic that refuses to lie about the unverified. Two-valuedness of the values does not mean classicality: the entailment relation is provably different (LEM fell, the deduction theorem is one-directional, $\neg\neg p \not\models p$ — §§3–5). A logic is defined by its entailment, not by its palette.

Local versus global reading of the mark. “Z is a mark on the atom” admits two readings of verdicts: the global one (“a formula is assertable if true under all substitutions into the marked atoms at once” — classical supervaluation) and the local one (“every operator consults the mark on the spot” — ZTL). The anchor cells separate them unambiguously: on $\neg\neg Z$ (ZTL: T, globally: F) and on $Z \leftrightarrow Z$ (ZTL: F, globally: T); the other five cells coincide. The global reading returns all classical tautologies but loses tabularity (supervaluation is not truth-functional) and both signature cells — the ladder of floors and the NaN signature “not equal to itself”. The anchors choose locality.

Why not four values. The temptation to include N as a fourth value (precedent: Codd’s two NULLs for RM/V2, rejected by industry) is declined: values are not multiplied, non-values are typed. N is the $\perp$ of the fixed-point iteration (§9), the pending of any promise: it exists only inside the solver and never returns outward. Read N as **Not-yet** — Kleene’s undefined *by intent*, at last housed as a phase rather than a value. The four-valued algebra {T,F,Z,N} with the lazy lift over ZTL is coherent and factorizes into our two registers ({T,F,Z} fragment = ZTL, {T,F,N} fragment = Kleene), but it packs two phases into one type — a monolith instead of modules; it is kept as a possible appendix, not as the core.

Kleene, read through this passport. Kleene’s third element was epistemic by intent — “undefined, not yet computed” — but his logic has a single register, so the status had no home except inside the value algebra, where

it was forced to flow ($\neg N = N$: negation passes the unknown on). Typed by our passport, his element conflates the two non-values: the *mark* on a datum (external, static, lifted only by the act of verification) and the *phase* of a computation (internal, dynamic, lifted by the iteration itself — and hardening, when it never resolves, into the quarantine mark with its own passport of kinds, §9). Strong Kleene logic is what results when both non-values are made to share one symbol *as a value of assertions*; SQL NULL’s notorious ambiguity — “unknown”, “not applicable”, “pending” in one symbol — is the same conflation observed in the wild. The fault is not the shared symbol (our own solver reuses Z positionally during iteration) but the promotion of a status to a truth value: ZTL splits the *role*, not the alphabet, and revokes the status’s right to be what a statement evaluates to.

A closing note on errors: the system has no error letter, and none is missing. An *error* is an interface event — the premature read of a phase: any value returned for a still-pending N is wrong whatever it is, because nothing has been computed yet. The engineering cousin pair makes the split visible: IEEE’s quiet NaN behaves like the mark Z (carried, tested, lifted by verification), while the signaling NaN is the alarm on touching the phase. The alphabet stays four letters; E is hardware, not logic.

Consequence for positioning: ZTL’s neighbours are not the many-valued logics but the two-valued assertability policies of the supervaluation family — from which it differs by locality, tabularity, and greedy collapse.

11. Expeditions: Curry, parity, Yablo, the crocodile (MEASURED)

Paradox as an operator — the expeditions are one construction (MEASURED, `pengine.py`). A self-referential net is a system $S_i = f_i(S_1 \dots S_n)$; to read it is to ground it (§9). So a paradox is $\text{paradox}(f) = \text{ground}(S = f(S))$ — feed an operator, form its self-reference, ground it — and the specimens below are that one construction on chosen operators: the liar is $\text{paradox}(\neg S)$, the truth-teller $\text{paradox}(S)$, Curry $\text{paradox}(S \rightarrow \perp)$, Russell’s propositional shadow again $\text{paradox}(\neg S)$. Three measured layers read a net, coarse to fine:

1. *Grounding* — the zero-trust verdict (§9): every pure self-reference lands in Z. Coarsest; it does not separate paradoxes.
2. *Solutions* — the classical models of the net: 0 = contradictory, ≥ 2 = underdetermined, 1 = determined. This is the content of the parity theorem below — an odd ring has no model $\Leftrightarrow$ its reference graph is not 2-colourable.
3. *Dynamics* — the period spectrum of the greedy jump: finest. The period-1 points are exactly the solutions, so dynamics refines (2); the higher periods separate what the model count merges — the liar and Curry are the *same* 2-cycle, the k-ring carries $2k$, the crocodile a 4-cycle.

Grounding is strictly the most conservative of the three: measured over all 9015 one-sentence nets up to six symbols, it reaches a classical value only when the net has a *unique* model that is also reachable from ignorance; 1068 of them have a unique classical model yet ground to Z (e.g. $S = S \vee \neg S$ — classically T, here Z: zero-trust will not grant a truth it cannot ground). The ZTL-settled nets are thus a strict subset of the classically-categorical ones.

Curry without negation. $c = (\text{Tr}(c) \rightarrow \perp)$ is the paradox that breaks naive paraconsistent theories (it uses no $\neg$, so taming negation does not help). Measured: no greedy fixed points, iteration of period 2, lazy grounding gives Z. **The same mechanics as the liar:** quarantine does not care which operator a sentence used to invert itself — only the non-existence of a fixed point matters. Our construction silences Curry for free, where LP must sacrifice contraction.

The parity theorem — total. All cycles of length 1–5, all inversion patterns (62 systems): classical models exist $\Leftrightarrow$ the number of inversions around the ring is even (the XOR-sum of the edges is 0). Odd rings are carousels (liar $n=1$, Jourdain $n=2$), even rings are truth-tellers (underdetermination).

Yablo: a third source of ungroundedness. $s_i = \text{“all } s_j, j > i, \text{ are false”}$ — a paradox without a single cycle. Measured: **every finite truncation is fully grounded** (the unique model $F \dots FT$, empty quarantine, exactly one greedy model). Yablo’s paradox lives only at actual infinity — a finite instrument cannot see it in principle. So there are three distinct sources of ungroundedness: the odd cycle (the liar), the odd infinite progression (Yablo), and the underdetermination of even structures (the truth-teller); the first is caught finitely, the second only in the limit.

The crocodile. “I shall return the child $\Leftrightarrow$ you predict what I will do”; the mother: “you will not return it”. Formalization: $R = \text{Tr}(M)$, $M = \neg \text{Tr}(R)$ — Jourdain’s carousel in disguise (cycle 2, one inversion, odd). MEASURED: no greedy models, iteration of period 4, mutual quarantine under lazy grounding, and — the key measurement — **the greedy verdict on the deal itself, $R \leftrightarrow M$, at the grounded point: F**. The zero-trust reading of the ancient dilemma: the deal’s condition cannot be grounded — the contract is void, no obligation ever arose; the crocodile is not “unable to comply” but “never contracted”. The control case (an optimistic mother, $M = \text{Tr}(R)$, an even cycle): classical models exist — (T,T) and (F,F), the word is kept in both — but the least fixed point is still quarantine and the deal’s verdict is still F: an enforceable self-referential contract does not self-enforce — which model realizes is decided by an external choice, not by logic. The difference between paradox and underdetermination is the difference between “the contract is void” and “the contract is valid but requires the parties’ will”.

12. Sets with unverified elements (MEASURED)

Sets are not postulated — they are derived: element equality is an atom (T/F for verified elements; Z whenever a mark is involved, including a mark with itself), membership is an $\exists$ -fold, inclusion an $\forall$ -fold, set equality mutual inclusion; everything computes through the core tables, with not a single special rule. Representation: (a core of verified elements, a quarantine multiset of marks).

Measured: $\{Z, Z\} \neq \{Z\}$ (merging is not earned — two unverified witnesses are not one); **$Z \notin \{Z\}$** even for the same mark (SQL: NULL IN (NULL) is not true); on clean sets the whole classical set theory is intact (C-extension); a mark falls exactly the **identity laws** — idempotence $S \cup S = S$ and $S \cap S = S$, self-subtraction $S \setminus S = \emptyset$, reflexivity $S = S$ and $S \subseteq S$ — the same families that fell in the logic: sets inherited the price list from the tables.

Cardinality is an interval: $|\{1, 2, Z\}| \in [2, 3]$, the exact value is not earned (verdict F); but $|\{Z\}| \in [1, 1]$ — **cardinality is earned even where identity is not:** one mark is exactly one thing. Cardinality and identity have split into different currencies of trust.

SQL’s inconsistency is not inherited: SQL holds $\text{NULL} \neq \text{NULL}$ in comparisons yet merges NULLs in DISTINCT/GROUP BY — swapping equality of values for equality of marks inside one syntax. Here the core deduplicates classically and the marks live with multiplicity — each operation honest about its own business.

13. The reals: two failures of enumeration (MEASURED)

A real-in-the-making is a stream of digits; at time t a prefix is verified. Stream equality is an atom with a pinned fate: prefixes diverged — **F earned** (apartness, a finite witness); they agree — **Z; T is never earned at any t** (the comparison is infinite). Finitely-presented objects (fractions p/q) are the contrast: equality decides finitely, the atoms are T/F.

Measured — uncountability splits into two distinct impossibilities:

1. **Non-registrability (zero-trust, about presentation).** Membership of a stream in any registry — an $\exists$ -fold of atoms from $\{F, Z\}$ — is eternally F: even streams literally standing in the list (including a duplicate!) earn no membership. No enumeration of streams can certify coverage of a single element —

including its own rows. Registries of fractions, by contrast, certify every element: the countability of $\mathbb{Q}$ is the earnability of presentation identity.

2. **Incompleteness (Cantorian, about cardinality).** The diagonal is an apartness-earning machine: against the i -th entry a finite witness is found by time $i+1$. The diagonal's non-membership is not postulated — it is earned.

Classical usage merges both impossibilities into the single word “uncountable”. The Z-optics splits them: the first fits into Z entirely (it is a property of extensional presentation and strikes even countable stream families), the second remains a cardinality fact which the diagonal renders *earned*. Resonance: the split “how many / which exactly” of §12 (cardinality earned without identity) is the same split seen sideways.

14. Functions: taint mode (MEASURED)

A function is a computation, not a verdict $\Rightarrow$ by the two-register theorem it behaves lazily: **the mark flows through the function** with a growing pedigree ($f(m)$ is a new mark “f applied to m”). Measured:

- **Images:** verified collisions *earn* merging ($f(1)=f(2)$ is a proven fact, the core deduplicates), marks keep multiplicity: $|f(\{1,2,3,Z\})| \in [2,3]$.
- **Composition:** taint is transitive (pedigree $g(f(m))$); the image is associative at representation level while verdict-equality is F (regularity R1, §24).
- **The preimage splits** into a verdict version (marks dropped — default deny) and a solver version (marks as candidates) — regularity R2 (§24).
- **The pearl: even the identity function is not certifiably injective on a marked domain** — pairs with a mark give Z-atoms, the implication $Z \rightarrow Z = F$, the $\forall$ -fold collapses. An injectivity certificate requires a fully verified domain; the echo of fallen $S \subseteq S$.
- **Laundering is forbidden:** functions do not remove marks; the only sanitizer is external verification of the value. In security terms: declassification only through proof.

The third engineering twin. After IEEE NaN and SQL NULL — **taint tracking / information flow control** (Denning's lattice 1976, Perl taint mode, TaintDroid): the Z-mark is taint, lazy flow through computations is taint propagation, greedy verdicts are sanitizer checks, the laundering ban is no-declassification.

15. Arithmetic with marks (MEASURED)

Numbers: verified values and marks with an interval of partial knowledge $[lo,hi]$ (ignorance = $(-\infty,\infty)$). Operations are computations $\Rightarrow$ lazy: intervals flow (interval arithmetic, decorrelated). Comparison atoms follow the generating principle extended to intervals: **T if forced under all readings; F if falsehood is forced; else Z.** Measured:

- **Forcedness earns even on marks:** $0 \cdot w =$ an earned 0 even for a wild mark (forced on $\mathbb{Z}$ by all readings) — a point of deliberate divergence from IEEE (their $0 \cdot NaN = NaN$: their domain contains inf/nan). $m - m \in [-9,9] \neq 0$ — decorrelation (like $NaN - NaN$, like $\{Z,Z\}$).
- **Three fates of an atom:** $[3,5] < [10,12]$ — T earned; $[3,5] = [10,12]$ — **apartness earned by intervals** (the echo of §13: difference is finitely witnessable); overlap — Z; the same mark against itself — Z (coincidence of bounds $\neq$ coincidence: identity is earned by nothing short of full verification $[x,x]$).
- **Verification = interval narrowing:** the atom “ $4 < m$ ” travels $Z \rightarrow Z \rightarrow T$ along $[0,9] \rightarrow [3,7] \rightarrow [5,7]$; what is earned is never revoked — the monotonicity of the lazy register, now in numbers.
- **Price-list inheritance:** commutativity of addition survives at the interval level, verdict-equality is $Z \rightarrow F$; the unit $x+0=x$ falls verdict-wise with coinciding intervals (regularity R1, §24).

The fourth twin: abstract interpretation (Cousot & Cousot, 1977) — interval value analysis (lazy flow of abstract values through computations) + assertion checking (greedy verdicts). Four independent engineering traditions — NaN, NULL, taint tracking, abstract interpretation — turn out to be fragments of one logic.

16. The probabilistic bridge: $Z \neq p = 0.5$ (MEASURED)

Three measurements answer how a mark of ignorance differs from a uniform prior.

Reparametrization (a discrete Bertrand). Given only $w \in [0,1]$; the question “ $w \leq 0.25$?”. A Bayesian with a uniform prior on w answers $1/4$; a Bayesian with a uniform prior on w^2 (the same ignorance!) answers $1/16$. One ignorance — two contradicting numbers: the choice of parametrization is smuggled information. The ZTL atom at $w \in [0,1]$ is Z in both parametrizations: **ignorance does not convert into a number without importing information**, and this is invariant.

Dempster–Shafer. On masses $m(\{a\})=m(\{a,b,c\})=1/2$ it is measured that the ZTL verdict of an event is the threshold of belief functions: $T \Leftrightarrow \text{Bel} = 1$ (forced by all readings of ignorance), $F \Leftrightarrow \text{Pl} = 0$ (excluded by all), else Z . The generating principle of ZTL is the $\{0,1\}$ -threshold of Dempster–Shafer theory; the interval cardinality of §12 is the elementwise [Bel-count, Pl-count].

Ellsberg. An urn with a verified 50/50 versus an urn of unknown composition: $\text{EV}(K)=[50,50]$ is an earned number, $\text{EV}(U)=[0,100]$ an interval; the atom “ U is no worse than K ” is not forced $\rightarrow$ default deny $\rightarrow$ choose K . The famous “irrationality” of Ellsberg’s subjects (1961) is the distinction between risk (verified p) and ignorance (a mark), inaccessible to a point prior: zero-trust rationality.

The second “SQL theorem” — about Bayesians. A point prior made from ignorance is a greedy laundering of Z into a number, the same substitution as merging NULLs in DISTINCT, and it is punishable (reparametrization). The honest architecture is two-registered: ignorance lives in mass intervals (the lazy register — Dempster–Shafer and **the fifth twin: Walley’s imprecise probabilities, 1991**), while decisions are verdicts by forcedness (the greedy one). Bayes remains honest on verified probabilities — his C -extension; only minting numbers out of emptiness is forbidden.

17. The modal layer: local $\Box$ versus global (MEASURED)

Worlds are the classical completions of the unverified atoms; $\Box\varphi$ = in all, $\Diamond\varphi$ = in at least one. Measured:

- **Atom verdicts are modal thresholds** (totally): $T \Leftrightarrow \Box p$, $F \Leftrightarrow \Box \neg p$, $Z \Leftrightarrow$ contingency; the duality $\Diamond = \neg \Box \neg$ holds; nested modality collapses ($\Box p$ is classical $\Rightarrow \Box \Box p = \Box p$) — the frame is S5-like. This is §16 one floor up: $\Box/\Diamond$ are the Bel/Pl thresholds in probabilistic dress.
- **The tableau signs are modal claims:** strict T/F are $\Box\varphi$ and $\Box\neg\varphi$, weak P/N are $\Diamond\varphi$ and $\Diamond\neg\varphi$. The calculus signature reads modally: proof demands necessity, refutation settles for possibility.
- **Three logics on one formula** (classical | global $\Box$ | ZTL): $p \rightarrow p$ and LEM: $T \mid T \mid F$ — supervaluation (one $\Box$ over the whole formula) preserves all classical tautologies, the local per-operator $\Box$ fells them. But $\neg\neg p$: $T \mid Z \mid T$ — **ZTL earns a verdict which global supervaluation cannot give** (the ladder of floors): the systems are incomparable, not ordered by strictness.

Result: **ZTL is a locally-modal logic over the S5 frame of completions**; every operator carries its own $\Box$ -collapse. Bochvar’s translations ($\rightarrow = \Diamond A \supset \Box B$, $\neg = \Box \neg$) acquire world semantics; the supervaluation/ZTL split turns out to be the global/local modality split. The theoretical relative is Hintikka’s epistemic S5 ($\Box$ = “known”): ZTL asserts only the known, but its modality is per-operator.

18. Russell: containment instead of explosion (MEASURED)

Russell is the liar dressed in membership: $R = \{x : x \notin x\} \Rightarrow R \in R \Leftrightarrow \neg(R \in R)$. The test universe: $a = \emptyset$, $b = \{b\}$ (a lawful eccentric), R ; a system of nine membership facts, with Russell's definition referring to the facts $x \in x$. Measured:

- **The greedy jump**: zero models, iteration of period 2, and exactly one cell oscillates — $R \in R$; the whole storm is localized.
- **Lazy grounding**: 8 of 9 facts grounded; $a \in R = T$ (does not contain itself — admitted), $b \in R = F$ (contains itself — rejected). **Russell's set works as a set for everyone except itself**; quarantine is one cell.
- **Verdicts**: “ $R \in R$?” — not earned, refusal; “ $R \notin R$?” — the greedy reading of $\neg(R \in R)$ is F — also refusal. The NaN signature reaches set theory: neither membership nor non-membership is earned.
- **The twin $S = \{x : x \in x\}$** : three greedy models (T, F, Z), lazy grounding Z — the truth-teller of set theory, underdetermination. Even and odd, now in sets.

The contrast with the classical outcome is maximal: for Frege one cell $R \in R$ blew up the whole system (from an asserted contradiction everything follows). In ZTL the same cell goes into quarantine and the rest of the universe stays grounded: **explosion requires an asserted contradiction, and quarantine asserts nothing**. The grounding is kernel-checked: the certified least-fixed-point iteration of §9 computes the nine-fact system to $[F, F, T, F, T, F, F, F, Z]$ — eight facts classical, quarantine exactly at $R \in R$ (Lean `russell_grounded`, zero axioms). A comparison with an earlier system of the author is instructive: in VR-Sets [Zenodo 10.5281/zenodo.20592428] Russell is excluded by grammar (forbidden to write), in ZTL he is admitted and defused pointwise. Two honest answers to one calamity: keep it out, or let it in under guard.

19. The verification operation and verdict warranties (MEASURED)

The act of verification — removing a mark and writing in the earned value — exposes a narrow place: **greedy verdicts are non-monotone under verification**. Not only refusals flip ($p \vee \neg p$: $F \rightarrow T$ upon $p := T$ — expected: default deny until checked) but **T flips too**: $\neg \neg p = T$ (a ladder report) dies at $p := F$. A verdict without a warranty is a Frege cell: an unfenced spot where a consumer who read T as “settled forever” builds on sand.

The fence is a warranty — and (corrected in v1.1) it is a ladder of two grades, not one bit.

- **The SOUND grade** (the stability bit of v1.0): all completions give one classical answer equal to the current greedy verdict — the global supervaluation of §17. It buys *never lies*: a sound verdict agrees with every possible resolution of the marks, so no truthful verification can ever reveal it to have been false. Cheap: one pass over the completions.
- **The HEREDITARY grade**: the verdict is unchanged under *every partial refinement* — any subset of the marks verified to any classical values. It buys *never spoils*: no verification path can revoke the verdict. Hereditary $\Rightarrow$ sound (completions are among the refinements); the converse is **false**.

Correction of the v1.0 claim (MEASURED). v1.0 asserted an equivalence — “stability-by-supervaluation $\Leftrightarrow$ invariance under every sequence of verifications (totally, 90 formula $\times$ marking pairs)” — and a monotonicity corollary. Both were facts about that 10-formula pool, not laws. The separating shape, found by the identity atoms of the operational-sets expedition (E21) on a 3303-formula pool and cross-checked with this section's own instruments, is **or(ladder, gap)**:

$\neg \neg p \vee (q \vee \neg q)$ — greedy T via the $\neg \neg$ ladder, insured by a gap that is true in ALL completions yet greedy- F ; verifying $p := F$ kicks the ladder away before the gap closes. The verdict was sound — and died. A simpler cell: $(\neg p) \rightarrow (q \rightarrow q)$, where the gap is the fallen law of identity itself.

Measured on the extended pool: hereditary-without-sound 0 (the ladder is real), sound-without-hereditary > 0 (the grades separate); the hereditary grade is monotone — never revoked, never degraded (0 violations, totally) — while sound-only verdicts are revocable. Subsequently stress-tested at scale (2026-07-12): a 151.8-million-pair hunt over four atoms at depth three, all six connectives, found 0 violations of the ladder inclusion and of hereditary monotonicity; the grade separation is generic (1.2 million cells), not exotic.

The fence depth is exactly $m-1$ (MEASURED). How deep must a verification-invariance check look before the hereditary grade is certain? For a *sound* verdict all full completions agree with it by definition, so heredity violations can live only at partial refinements of size at most $m-1$ (m the number of marks): depth $m-1$ always suffices. It is also necessary: the guard family

$$(b_1 \wedge \dots \wedge b_{m-1}) \rightarrow (a \rightarrow a)$$

— a conjunction guard of $m-1$ marks standing over the fallen law of identity — is sound, invariant under every verification of fewer than $m-1$ atoms, and dies the moment all guards are verified true, when the door opens onto the greedy-F gap $a \rightarrow a$ (checked deterministically for $m = 3, 4, 5$; the $m = 2$ witness is the cell $(\neg p) \rightarrow (q \rightarrow q)$ above). Hence **no constant-depth characterization of the hereditary grade exists**; the cost of the full warranty grows with the number of unverified inputs, and what remains open is a structural, non-enumerative criterion.

Result: a verdict is a pair (value, warranty grade). The value is greedy (local, fast); the warranty grades are global. Six verdict classes now, and the honest advice differs by grade: **hereditary T** — **build your house** (no verification path can revoke it); **sound T** — **never a lie, but may stall to refusal before verification completes**; T-until-verification — a ladder report $(\neg \neg p)$, alive till the first check; symmetrically for F, with F-until-verification as default deny. The Frege cell is fenced by the top grade only; the middle grade fences lying, not spoiling.

20. Evidence combination: conflict is not laundered (MEASURED)

Pieces of evidence about one value are constraints; **combination = intersection**. Measured:

- **The unification theorem:** verify of §19 is a special case of combination (a singleton witness $[v, v]$); verification and evidence fusion are one operation. An honest side effect: the act of checking can itself earn a conflict (verify 7 against $m \in [0, 5]$ — the checker against the prior evidence).
- **An empty intersection is an earned contradiction of sources** (a sound F for the verdict “both are honest”), not noise for renormalization.
- **Zadeh’s paradox (1984), resolved in Smets’ favor.** Two doctors almost exclude a tumor (0.01 each); Dempster’s rule renormalizes the 0.9999 conflict away and outputs “tumor = 1” — an unshakable false certainty. Smets’ conjunctive rule (TBM) keeps the conflict in $m(\emptyset)$ — our approach: the conflict is exhibited, the diagnostic verdict is refusal until the doctors are sorted out. **Renormalizing conflict is the same laundering of ignorance as the uniform prior (§16), in the chapter on combination: one principle, two diseases.**
- **The sixth twin: provenance polynomials (Green–Karvounarakis–Tannen, 2007).** The pedigrees of marks (§14) grow into polynomials over sources: a fact with derivations $A \cdot B + C$ stays alive while at least one monomial lives; retracting a source zeroes a variable. An algebra of trust in derivations, measured on retraction scenarios.

The twin count: **six** — NaN, NULL, taint/IFC, abstract interpretation, imprecise probabilities, semiring provenance.

21. Logical time: verification is the only clock (MEASURED + Lean)

ZTL owns no physical clock, and §7 keeps it honest: duration, tense and “how long” are outside the system’s axis. Yet one clock was inside the calculus all along, unnamed. The verification operation of §19 is a tick: one act *verify* resolves one mark into an earned classical value. A moment is a marking; the past is the verified prefix; the future is a tree — every remaining mark can resolve either way, so time branches, and “everyone’s time differs” is just the choice of a path through one tree.

The ladder is a temporal logic. The central identification costs nothing and buys the whole layer: the warranty ladder of §19, read on the verification tree, is a system of temporal quantifiers —

- *until-verification* — true **now** (the present tense of a verdict; credit);
- *sound* — true **at every ending** (all completed traces agree; the road may wobble);
- *hereditary* — true **always along every path** (the invariant of the tree; never revoked).

The modal layer of §17 is the statics of this structure (worlds = completions); the temporal layer is its dynamics — the paths through the partial refinements between here and the endings.

Measured (`ztime.py`; the grade automaton over the exhaustive depth- ≤ 2 pool, 2,906 formulas, 29,812 ticks): the hereditary states absorb — not one tick leaves them (0 violations); no compound formula is ever caught waiting (greediness in temporal costume); ground can arrive all at once (14,818 direct $U \rightarrow H$ jumps) and credit can worsen before it settles (108 $S \rightarrow U$ demotions); every completed trace ends hereditary — the arrow of logical time points at the shelf (130 of 130); and settlement can come early — 68 of 130 traces reach a hereditary verdict with marks still unresolved (after $p := T$ the verdict of $p \vee q$ no longer cares about q). Settling times run from 0 ($\neg(p \wedge \neg p)$ is born on the shelf) to every-mark ($\oplus$ needs them all).

Lean (`lean/ZTime.lean`, empty axiom list, structural — for every formula, marking and tick, not an enumeration): absorption (a hereditary verdict survives any tick with its grade), the arrow (a fully verified marking is hereditary), and the ladder inclusion (hereditary $\Rightarrow$ sound; an ending is a refinement). The proofs are pointwise throughout — no function extensionality enters.

A conjecture that lived one hour — kept, as the method demands. The first sweeps found no tick that ever *enters* the sound-only grade, and for an hour the conjecture stood: *sound is a birth grade — verification spends it but cannot mint it*. The proof attempt broke at a specific spot, and the spot folded into a counterexample: the pools had been shallow. The selector $\varphi = (a \wedge X) \vee (\neg a \wedge p)$ with $X = \neg p \vee (q \vee \neg q)$ walks $F/U \rightarrow (a := T) \rightarrow T/S \rightarrow (p := T) \rightarrow T/H$ from the all-marked start: soundness is *earned* — by the tick that verifies which world you are in — and the full strict ladder $U \rightarrow S \rightarrow H$ is realized rung by rung (`ztime.py` §§5–6, the record of the death included).

The tool. The ZFL statement genre carries an optional `timeline` field (a list of verification ticks, validated with its own error codes); the engine plays it into a chronicle — verdict and warranty per tick, with settlement marked. The consumer sentence of the layer: once a verdict is hereditary, every remaining check buys nothing — stop paying (`usage/car.py`, the used-car stand: an affirmation is earned at the last tick; a refusal is grounded at the first failure, three checks saved; verifying the selector first saves two).

22. Epochs: expiry and the boundary theorem (MEASURED + Lean)

The time of §21 is monotone: ground only arrives. Institutions live in non-monotone time — confirmations lapse, registries are re-pledged, facts are authoritatively changed. The anti-tick *expire* returns an earned value to the mark, and with it the layer splits into two chronologies that must never be conflated: the *knowledge* chronology (verify — learning more about the same world) and the *validity* chronology (expire — the world becoming different). A maximal verify-only stretch is an **epoch**; an expiry opens a new one. A composite event

— authoritative revocation — is deliberately a composition, expire + verify of the new value, so that one event cannot hide two institutionally different facts.

Measured (`zexpire.py`): hereditary is a warranty against future *verification*, not against the *loss of ground* — a fully verified conjunction (T/hereditary) falls to F/until-verification on a single expiry; and the “checks saved” of §21’s early settlement are revealed as a loan against the expirable ground: the settled verdict survives the death of its shortcut iff the saved checks were verified before the clock ran out — expiry-insurance, priced by the core.

The Epoch Boundary Theorem (`lean/EpochBoundary.lean`, empty axiom list, structural). Call a verdict *epoch-blind* at a marking if it is invariant along every finite chain of verify and expire events — i.e. if the protocol refuses to distinguish epistemic refinement from validity change. Then, within this model (markings over {T, F, Z}, the greedy evaluation, the two event kinds):

a verdict is epoch-blind **iff** it is constant over all markings whatsoever — iff the assertion reads none of its grounds.

Contentful conclusions cannot survive unrestricted epoch crossing; the empirical census (2,906 formulas, 0 contentful survivors) is thereby upgraded from evidence to a theorem for every formula of the language. A separation witness (`epochs_matter`) completes the picture: a hereditary verdict destroyed by one expiry — so intra-epoch warranty and cross-epoch persistence are provably different notions, and the epoch boundary is a logical necessity, not an administrative convenience.

The institutional reading. For an admission formula over grounding atoms the layer yields a three-coordinate cell — epistemic status (the global supervaluation of §17: Z is *not established* and is never conflated with falsity), operational decision (the greedy verdict: default deny), and warranty grade — and an event ledger in which every verification tick carries its source. The closed-world loan — “no proof of revocation, hence not revoked” — cannot be taken inside the logic at all: $\neg R$ at $R = Z$ evaluates F/until-verification, an argument from absence never yields T. It can enter only as a tick without a source, which the ledger exposes as an *ungrounded verification event*. A worked, frozen artifact (admission condition; expiry branch; revocation branch; the rejected ungrounded tick) ships in the repository (`vrg/epoch_artifact.py`, deterministic JSON ledger).

23. The price of derivations: transport, not creation (MEASURED)

The price list of §3.1 concerns laws; it extends to *paths*. Take the 12 alive entailment rules of §3.2 as the only links, the 2 fallen ones ($\neg\neg$ -elimination and tautology-in-conclusion) as a loan library, and close premise sets under chains, bounded to a 153-formula pool over three atoms; every closure is cross-checked against the semantic entailment of §3.2 (0 violations — the chains never lie; `zderive.py`). Three facts emerge.

From nothing, nothing — even on credit. The closure of the empty premise set is empty, with and without the loans: the battery has no axiom rule. And yet ZTL-tautologies *exist* — the guarded forms $\neg q \rightarrow \neg q$ and $\neg(p \oplus p)$ are true under every assignment, because denial is classical ($\neg Z = F$; §3.1’s “a denial is free”) — and the battery cannot mint even those. The alive rules are *transport, not creation*: classical logic mints truth from form; ZTL’s free truths must *enter*, as verified premises.

The one-way street, priced. From {p} the ladder $\neg\neg p$ is earned ($\neg\neg$ -introduction is alive); from $\{\neg\neg p\}$ the alive closure is $\{\neg\neg p\}$ itself, and the $\neg\neg$ -elimination loan unlocks fourteen formulas — the first measured *on-credit derivations*, p among them. What classical logic cannot even see (the step is invisible inside it) is here a priced borrowing with a named creditor.

The gap, honestly. The rules are incomplete for the semantic entailment even on the small pool ({p}: 24 entailed, 15 derivable) — part pool-boundedness, part missing law-rewrite links, part genuine; the split needs a

bigger pool, not a claim.

24. Cross-cutting regularities

Three facts recurred in every applied chapter; we fix them once.

R1. Two levels of equality. On every floor (sets, functions, arithmetic) the operations coincide at the representation level (the machine sees equality), yet verdict-equality refuses: the very act of recognizing identity is default deny. The engineering level serves the solver, the verdict level serves assertions.

R2. Two-registeredness reproduces itself. The verdict and solver variants of each construction (preimages, cardinalities, probabilities) arise without special design — as consequences of the generating principle and the §9 theorem on the necessity of two registers.

R3. Apartness is earned, identity is not. The difference of two unverified objects is earned by a finite witness (diverged intervals, diverged prefixes); identity is earned by nothing short of full verification. Hence in one stroke: $\{Z, Z\} \neq \{Z\}$, $m-m \neq 0$, the non-registrability of streams (§13), and the NaN signature $x \neq x$.

25. Roadmap

A Lean port of the parameter (arbitrary-domain) tableaux of §6; a syntactic cut-elimination procedure with complexity bounds (admissibility is settled — §5); the mining of the equivalent quasivariety scouted in §3.7 (axiomatization, subquasivariety lattice, a representation theorem replacing Plonka sums — a separate work); a cheap characterization of the hereditary warranty grade of §19 (sound is one pass over completions; is hereditary computable without enumerating refinements? — MEASURED narrowing, 2026-07-12/13: the fence depth is exactly $m-1$ in the number of marks — sufficient for every sound verdict (violations cannot hide in full completions) and necessary by the guard family $(b_1 \wedge \dots \wedge b_{m-1}) \rightarrow (a \rightarrow a)$, checked at $m = 2, 3, 4, 5$ (`zverify` §§5–6); hence NO constant-depth characterization exists and what remains open is a structural, non-enumerative criterion); a practical zero-trust validation library (verdicts with warranties + evidence combination + provenance); temporal operators over the verification tree ($\Box/\Diamond$ /until against the grade semantics of §21) once a concrete user forces them, and the grade automaton at scale; The interactive studio — natural language negotiated into ZFL and judged by the measured engines — already ships in the repository (`τool`); a possible essay, “Reinventing Bochvar through NaN”, remains on the horizon.

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Acknowledgements and AI disclosure

This work was carried out with the substantial participation of the AI system Claude (Anthropic) in a dialogue setting: the system generated the text, the test-bench code, and the Lean proofs. Across versions the dialogue ran on two Claude models, and the attribution is kept honest: Claude Opus 4.8 — the original corpus (v1.0) and the 2026-07-14/15 additions (the three-laws capstone §3.1, the Finn attribution and reconciliation in §3.8, the paradox-engine synthesis opening §11, the Łukasiewicz pedigree and the genetic order in §4); Claude Fable 5 — the v1.1 same-day correction and the v1.2 assembly (the census of sixteen and its Lean clone equalities, the fence-depth theorem, the warranty ladder at scale, the naming of the lift, the temporal layer §§21–23 with `ZTime.lean` and `EpochBoundary.lean`, the expiry and derivation stands, and this PDF build). All design decisions, fork choices, hypotheses, and the final responsibility for the content rest with the human author. In accordance with COPE/ICMJE recommendations, the AI system is not listed as an author. The reliability of the results does not depend on trusting the AI: every numerical claim is checkable by the repository code (`run_all.py` — full regression), and every Lean claim by the Lean 4 kernel (empty axiom list, `#print axioms`).